

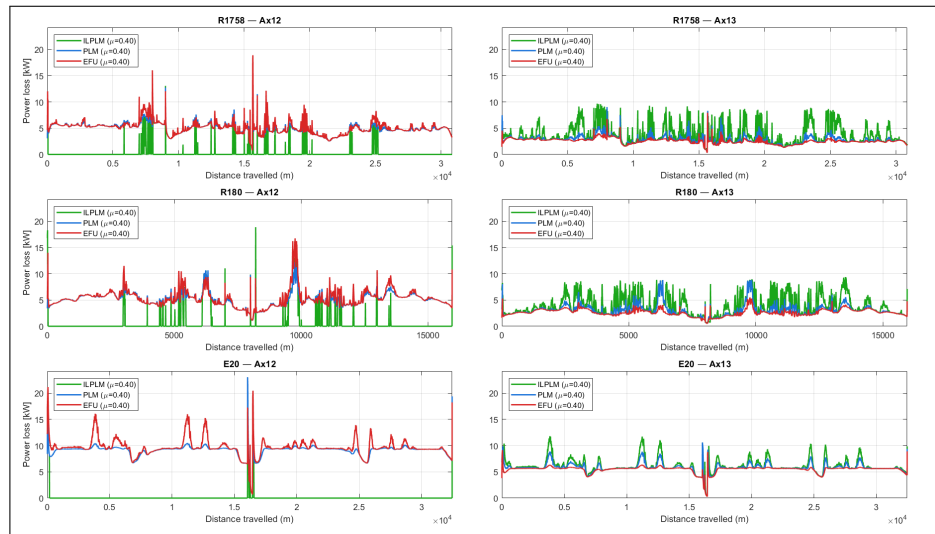


Figure 5.25: Axle Switching (6x4 rigid, $\mu = 0.40$)

6

Conclusion

For one to conclude, the main bargain comes in whether to choose the most efficient strategy or the safest strategy. Thus, the optimal solution is to find a solution that is energy efficient as well as safe.

Figures 6.1 and 6.2 highlight the trade-off between the lateral stability and normalised total energy consumed for the three different wheel torque coordination strategies. The comparison is established based on the value of the 20th percentile (or 20% of the total number of samples) corresponding to the lateral index margin (index 3) for a 4x4 truck.

The normalised total energy consumed is defined as the sum of the total energy losses (rolling resistance, longitudinal tyre slip, drivetrain, friction brakes) normalised with the product of the distance travelled by the vehicle during the simulation, gross weight of the vehicle, and acceleration due to gravity:

$$\text{Normalised Total Energy Consumed} = \frac{\text{Total Consumed Energy}}{\text{Distance travelled} \times GW_{\text{truck}} \times g} \quad (6.1)$$

where, Normalised total energy consumption: (dimensionless), Total energy consumed: [J or N · m], Distance travelled: [m], GW_{truck} : Gross weight of the truck [kg], g : Gravitational constant [m/s^2]

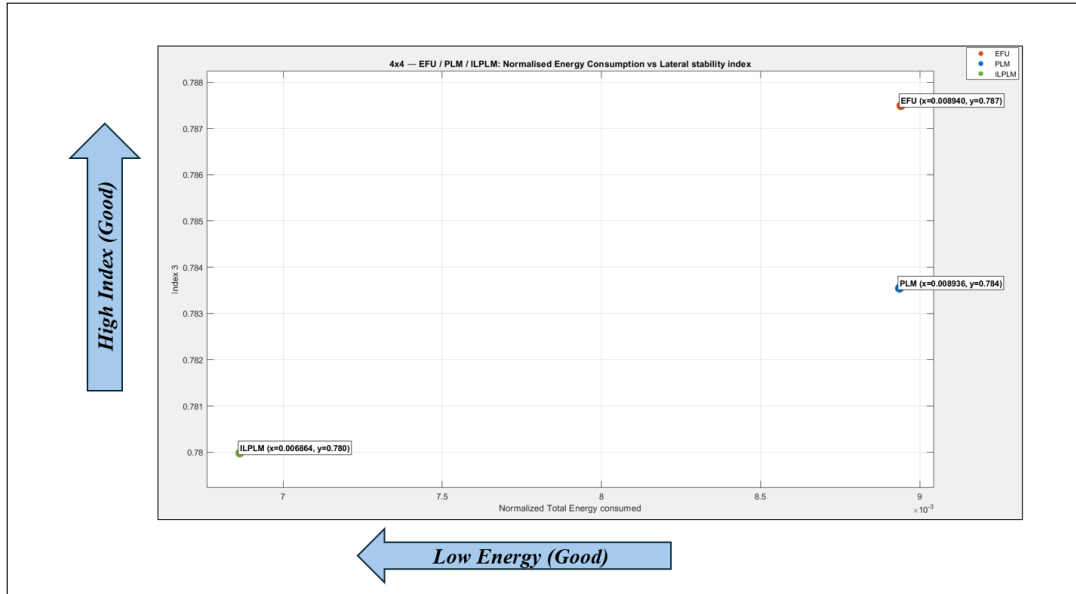


Figure 6.1: Trade off plot(4x4 rigid $\mu = 0.40$)

Figure 6.2 highlights the case for a 6x4 rigid truck and uses the same principle as used above. The only modification done is by replacing the index 3 with index 4 here.

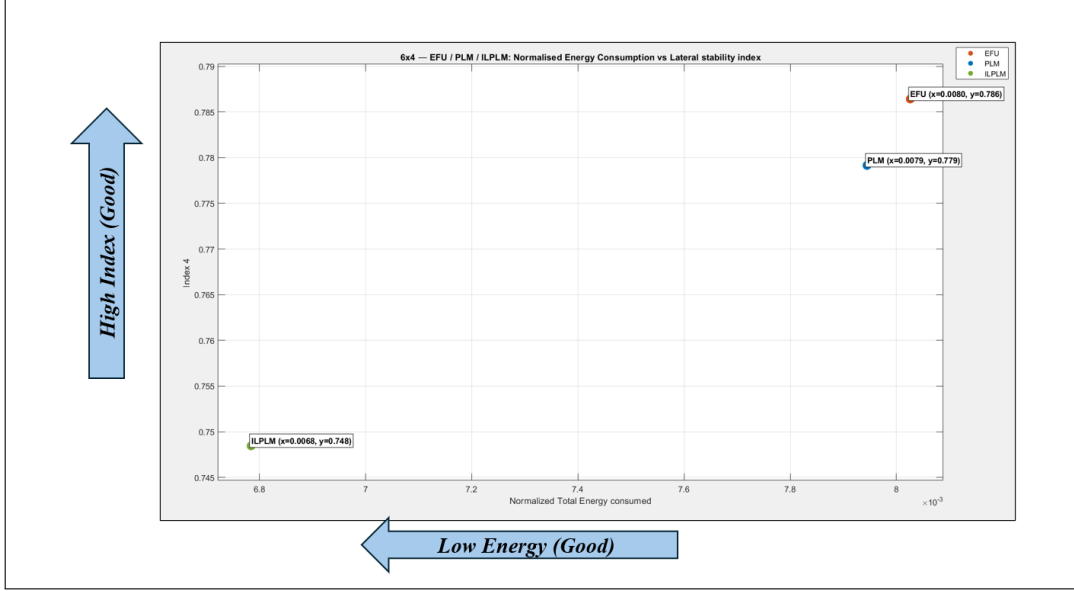


Figure 6.2: Trade off plot(6x4 rigid $\mu = 0.40$)

To comprehend the trade-off plots, as lower energy is consumed (enhanced efficiency), the lateral margin index tends to decrease slightly. Thus, EFU exhibits high stability (higher value of lateral index) with a higher value of normalised total energy consumed, followed by PLM. While ILPLM depicts higher energy savings, but with less lateral margin index.

Thus, to conclude the research, the following objectives have been fulfilled:

- Proposed metrics for vehicle stability and energy savings for different truck configurations with different axle positions for comparing all the wheel torque coordination strategies.
- The energy savings have been evaluated for the wheel torque coordination strategies for different kinds of configurations, which have proven to be the best in ILPLM, followed by PLM and EFU for both 4x4 and 6x4 rigid trucks.
- In terms of lateral stability, EFU is proven to be more stable, followed by PLM and ILPLM in case of both 4x4 and 6x4 rigid trucks.

7

Future Work

- Improving the Driver model to reduce aggressive driving behaviour.
- To make a longitudinal wheel slip controller for lower friction roads.
- Investigation of City Driving Scenario.
- Study on Construction Site Driving Scenario.

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Appendix 1

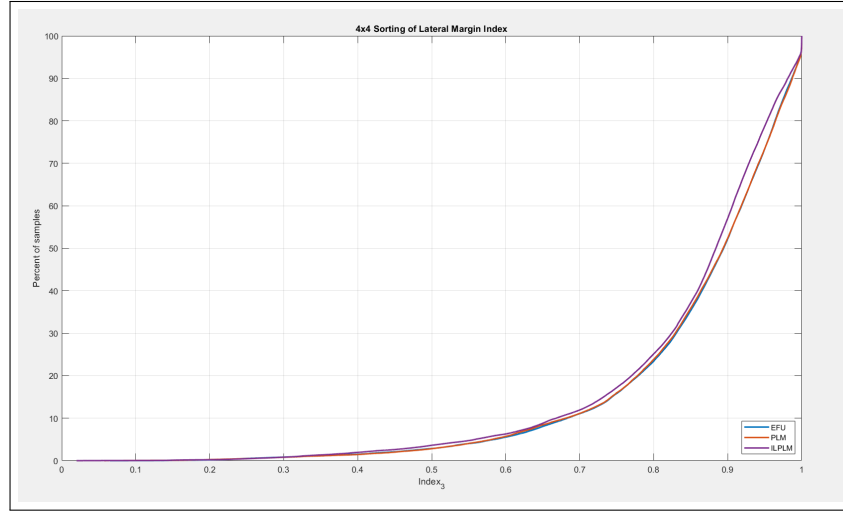


Figure A.1: Percent of samples versus index (4x4 rigid, $\mu = 0.40$, R180)

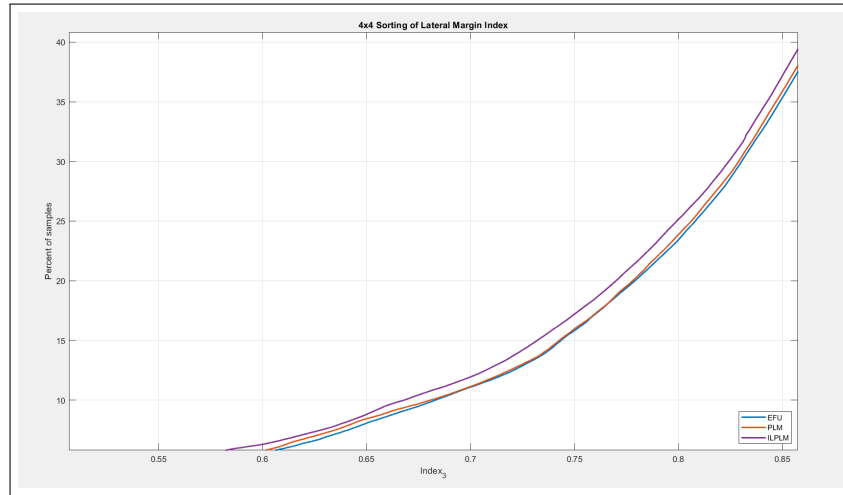


Figure A.2: Percent of samples versus index, Zoomed view (4x4 rigid, $\mu = 0.40$, R180)

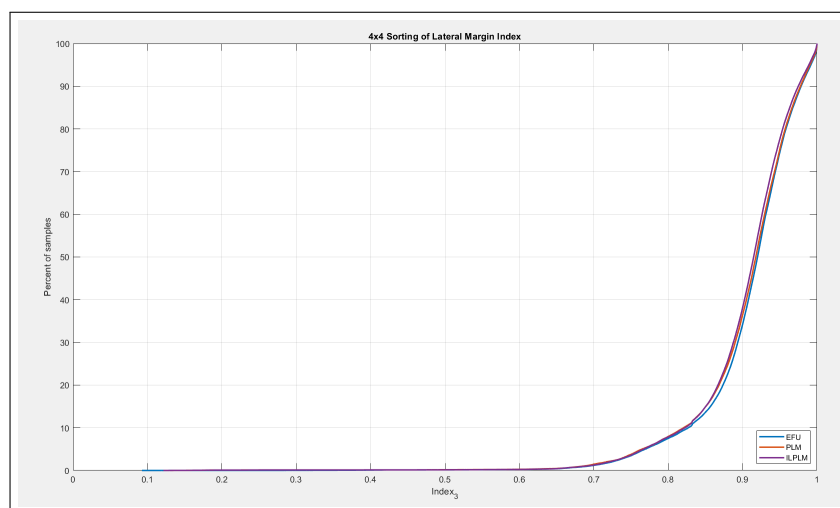


Figure A.3: Percent of samples versus index (4x4 rigid, $\mu = 0.40$, E20)

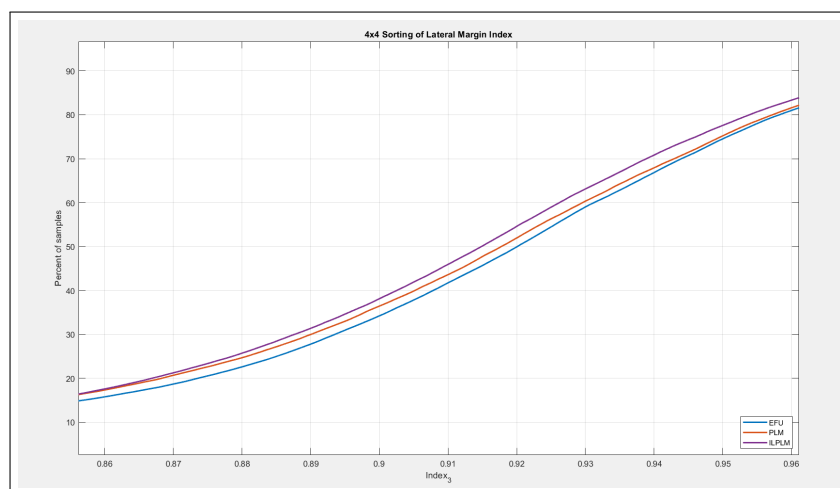


Figure A.4: Percent of samples versus index, Zoomed view (4x4 rigid, $\mu = 0.40$, E20)

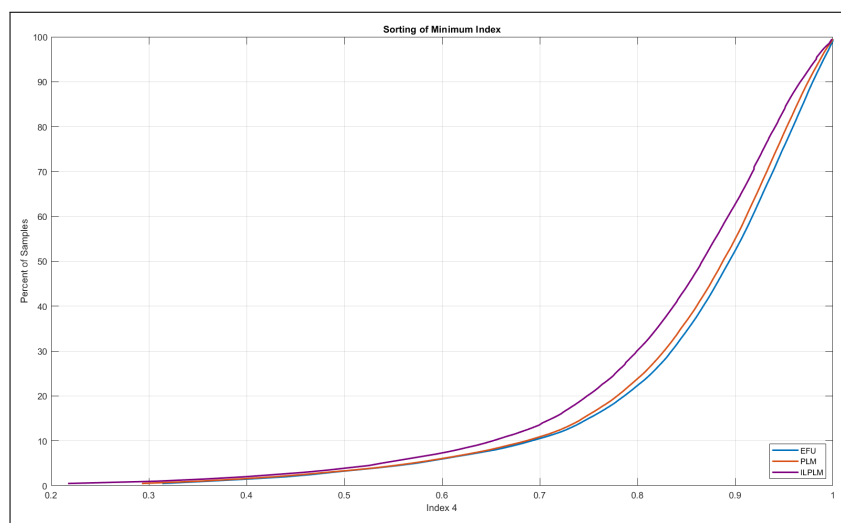


Figure A.5: Percent of samples versus index (6x4 rigid, $\mu = 0.40$, R1758)

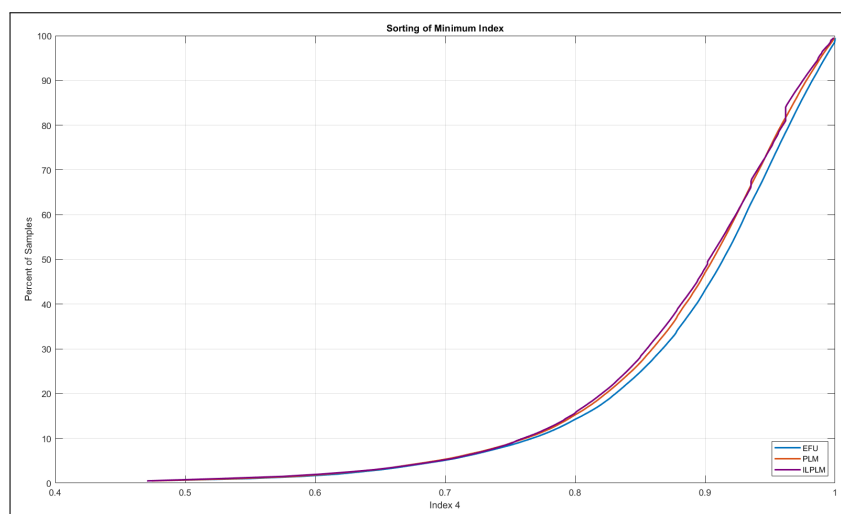


Figure A.6: Percent of samples versus index (6x4 rigid, $\mu = 0.40$, E20)

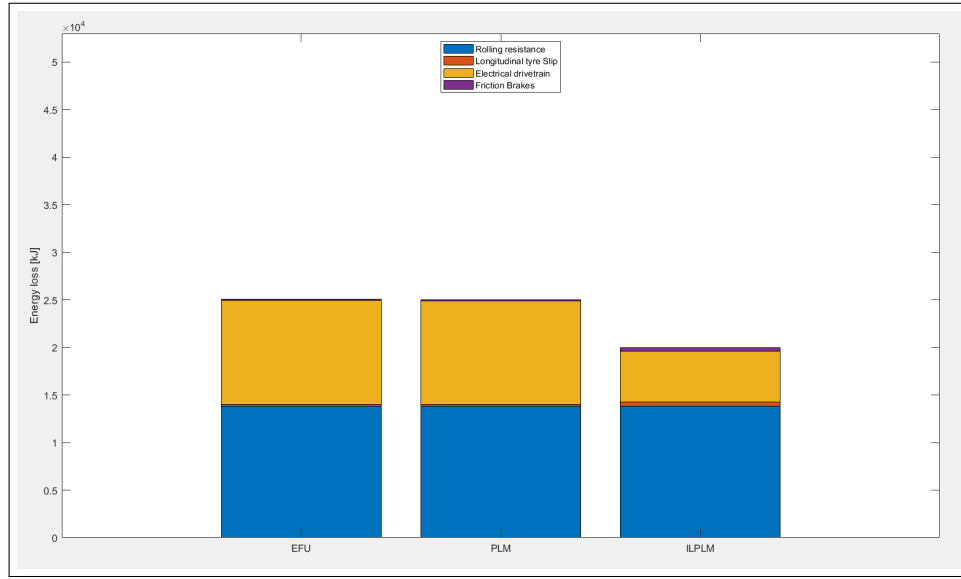


Figure A.7: Energy plot (4x4 rigid, $\mu = 0.40$, $R180$)

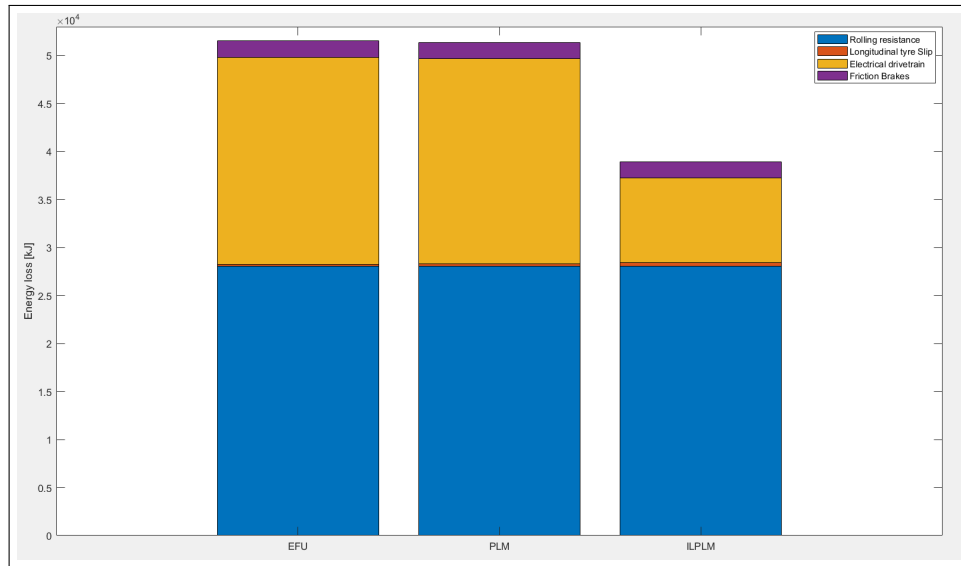


Figure A.8: Energy plot (4x4 rigid, $\mu = 0.40$, $E20$)

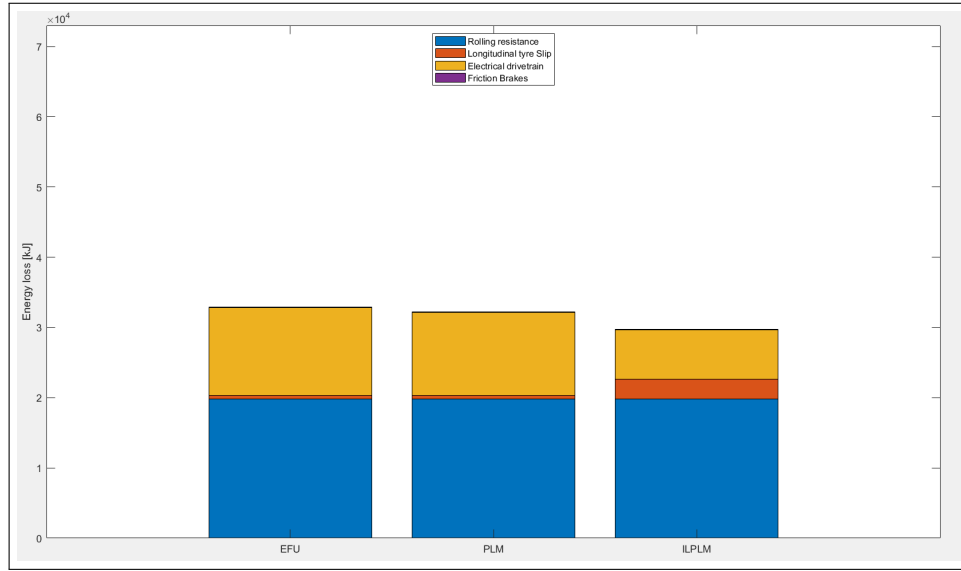


Figure A.9: Energy plot (6x4 rigid, $\mu = 0.40$, R180)

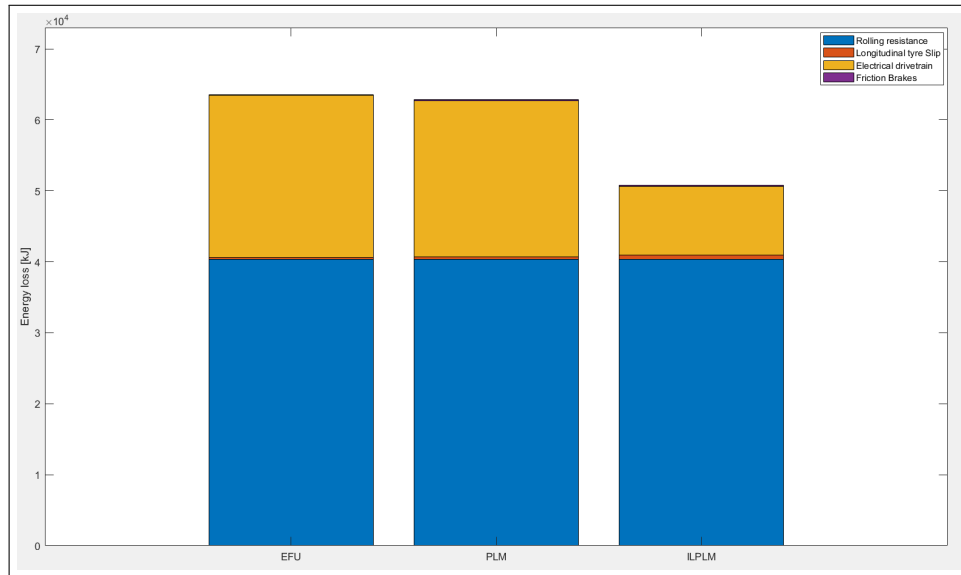


Figure A.10: Energy plot (6x4 rigid, $\mu = 0.40$, E20)

DEPARTMENT OF MECHANICS AND MARITIME SCIENCES

CHALMERS UNIVERSITY OF TECHNOLOGY

Gothenburg, Sweden 2025

www.chalmers.se



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